Lab Week -8

Huffman Coding

" Data Analytics And Intelligence Laboratory"

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"DATAANALYTICSANDINTELLIGENCELABORATORY"

Frequency:

| Letter | Count |
|---|---|
| A | 7 |
| L | 4 |
| N | 4 |
| T | 4 |
| E | 3 |
| I | 3 |
| C | 2 |
| D | 2 |
| O | 2 |
| R | 2 |
| Y | 2 |
| B | 1 |
| G | 1 |
| S | 1 |

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| B | G | S | C | D | O | R | Y | E | I |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 3 | 3 |

| L | N | T | A |
|---|---|---|---|
| 4 | 4 | 4 | 7 |

Applying huffman algorithm :

B[1] - G[1] - S[1] - C[2] - D[2] - O[2] - R[2] - Y[2] - E[3] - I[3] - L[4] - N[4] - T[4] - A[7]

I-1 Combine B[1] and G[1] to get BG[2]

S[1] - C[2] - D[2] - O[2] - R[2] - Y[2] - BG[2] - E[3] - I[3] - L[4] - N[4] - T[4] - A[7]

I-2 Combine S[1] and C[2] to get SC[3].

O[2] - D[2] - R[2] - Y[2] - BG[2] - E[3] - I[3] - SC[3] - L[4] - N[4] - T[4] - A[7]

I-3 Combine D[2] and O[2] to get DO[4]

R[2] - Y[2] - BG[2] - E[3] - I[3] - SC[3] - L[4] - N[4] - T[4] - DO[4] - A[7]

I-4 : Combine R[2] and Y[2] to get RY[4]

BG[2] - E[3] - I[3] - SC[3] - L[4] - N[4] - T[4] - DO[4] - RY[4] - A[7]

I-5: Combine BG[2] and E[3] to get BGE[5]

I[3] - SC[3] - L[4] - N[4] - T[4] - DO[4] - RY[4] - BGE[5] - A[7]

I-6: Combine I[3] & SC[3] to get ISC[6]

L[4] - N[4] - T[4] - DO[4] - RY[4] - BGE[5] - ISC[6] - A[7]

I-7: Combine L[4] and N[4] to get LN[8]

T[4] - DO[4] - RY[4] - BGE[5] - ISC[6] - A[7] - LN[8]

I-8: Combine T[4] & DO[4] to get TDO[8]

RY[4] - BGE[5] - ISC[6] - A[7] - LN[8] - TDO[8]

I-9: Combine RY[4] and BGE[5] to get RYBGE[9]

ISC[6] - A[7] - LN[8] - TDO[8] - RYBGE[9]

Iteration 10: Combine ISC[6] and A[7] to get

ISCA[13]

LN[8] - TDO[8] - RYBGE[9] - ISAC[13]

I-11: Combine LN[8] & TDO[8] to get LNTDO[16]

RYBGE[9] - ISCA[13] - LNTDO[16]

I-12 Combine RYBGE[9] & ISCA[13] to get

RYBGEISCA[22]

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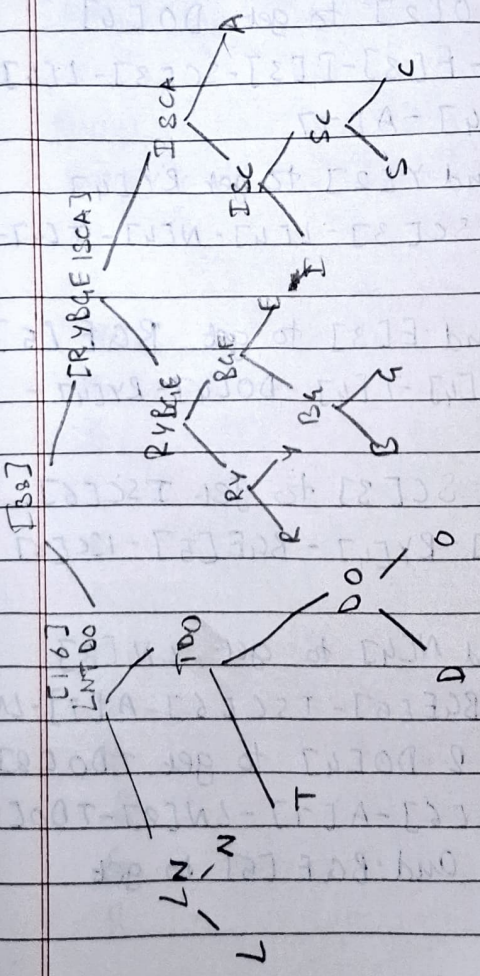
LNTDO[16] - RYBGEISCA[22]

Combining the

I-13:

LNTDORYBGEISCA[32]



A = 111 ; L = 000 ; N = 001 ; T = 010 ; E = 1011 ;
I = 1100 ; C = 11011 ; D = 0110 ; O = 0111
R = 1000 ; Y = 1001 ; B = 10100 ; G = 10101 ;
S = 11010